MIDDLE EAST TECHNICAL UNIVERSITY
Department of Electrical and Electronics Engineering

EE 301 Signals & Systems

Homework 2

Due: November 4, 2017

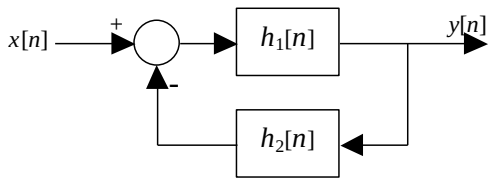
Problem 1: Prove whether each of the following statements is TRUE or FALSE.

a) It is possible that a causal CT LTI system produces positive output *y(t)* for some *t<0,* in response to a given input signal.
b) A nonlinear time-invariant system may not be uniquely described by its response to an impulse.
c) Two LTI systems, S_1 and S_2 are connected in cascade. If the overall system S is causal, then S_1 is also causal.
d) The system described by *y(t)=x*(t)* is LTI.
e) The system described by *y(t)=x(-t)* is LTI.
f) $y(t)=x(t)*h(t)$ implies $x(t-t_0)*h(t-t_0) = y(t-2t_0)$.
g) An LTI system may produce a nonzero output when the input is zero for all t.

Problem 2: Consider the feedback connection of two DT LTI systems as shown, where

$$h_1[n] = \delta[n], h_2[n] = (-3)^n u[n-1]$$

Determine the impulse response *h*[n] of the interconnected system assuming that the system is initially rest. Are the systems represented by $h_1[n]$ and $h_2[n]$ stable? Find the input-output relationship of the overall system. Is the overall system stable? Is it causal?



Problem 3: Let $x(t) = u(t-3)$ and $h(t) = e^{-3t}u(t)$. Compute $y(t) = x(t) * h(t)$, and $g(t) = \frac{dx(t)}{dt} * h(t)$. How is $g(t)$ related to $y(t)$?

Problem 4: Consider the causal LTI systems described by the differential equations:

a) $$\frac{d^2}{dt^2}y(t) - 2\frac{d}{dt}y(t) + y(t) = x(t)$$

b) $$3\frac{d^2}{dt^2}y(t) + 6\frac{d}{dt}y(t) + 3y(t) = 3\frac{dx(t)}{dt}$$

c) $$\frac{d^2}{dt^2}y(t) + \frac{d}{dt}y(t) - 2y(t) = 5x(t-1)$$

For each system, find the impulse response and determine whether it is stable.

Problem 5: Consider the causal LTI systems described by the difference equations:

a) $y[n]+\frac{1}{2}y[n-1]=x[n]$

b) $y[n]+\frac{1}{2}y[n-1]=3x[n-2]$

c) $y[n+2]+\frac{1}{2}y[n+1]=3x[n]$

d) $y[n]+4y[n-1]+4y\left[n-2\right]=x[n]$

Find the impulse response for each of the above systems and determine whether it is stable.